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NO TECTONIC ROTATION FOR THE APULIA-GARGANO FORELAND IN THE PLEISTOCENE

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Abstract. In Apulia (southern Italy), 11 quarries in late Pliocene and early Pleistocene clays were sampled to study the rotation of the underlying Apulian Unit. To analyze the Natural Remanent Magnetization (NRM) of the samples thermal magnetization was performed. Several NRM-components were distinguished. The component removed between 350-380°C and 450-480°C is the most consistent one; it also shows a positive reversal test. There is no rotation for this so-called HT-component. Mean inclination is the same as the inclination of the geocentric axial dipole (GAD) field for the present latitude of the area. It is concluded that these young clays reflect the geomagnetic field quite clear and that this part of the Adriatic microplate did not have any tectonic rotations since the Pleistocene.

Introduction

In southern Italy, the area between Salerno and Bari is divided into three main tectonic units: the southern Apenninic belt, the Bradano unit and the Apulia-Gargano foreland (Figure 1). The most external one, the Apulia-Gargano foreland is part of the Adriatic microplate; it has a continental basement covered by a carbonate succession of Mesozoic age: the Apulia Unit. This Apulia Unit extends below the Bradano Unit. The Bradano Unit represents a terrigenous succession, several 1000 m thick, accumulated in the Plio-Pleistocene foredeep of the southern Apenninic belt. The Bradano Unit as well as the underlying Apulian Unit are only affected by some normal faulting. Compressive structures only occur in the western part of the Bradano Unit, in connection with the southern Apenninic belt.

Paleomagnetic studies on rocks of 'pre-Neogene' age on the Apulia-Gargano foreland, especially Cretaceous limestones from Gargano [Channell and Tarling, 1975; Channell, 1977; VandenBerg, 1983] revealed a characteristic mean direction implying a counter-clockwise (CCW) rotation of 30° since the early Tertiary (relative to the magnetic north). Until recently only one paleomagnetic study in this area deals with rocks of Neogene age; Tozzi et al. [1988] report a possible clockwise (CW) rotation (25°) of southern Apulia (Salento) since the Eo-Oligocene. Their results imply that either the CCW rotation measured in the Cretaceous rocks has been compansated by the more recent CW rotation or that a major discontinuity decoupled the Gargano from the Salento during late Tertiary times. The results from Plio-Pleistocene sediments in that study did not give reliable results; sites in Pleistocene sediments from the southern Apennines show a CCW rotation of 25-30° [Sagnotti, 1992]. The aim of this study is to compare the results from Apulian Plio-Pleistocene sediments with the known paleomagnetic data.

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Sampling and laboratory treatment

The present study deals with (marly) clays overlying the Apulian Unit; the clays are late Pliocene and early Pleistocene in age. In 11 quarries, sites were selected on lithological criteria; fine-grained levels were sampled because of their suitable magnetic properties and because their sedimentation rate is usually low enough to average out, to a large extent, secular variation. In 5 different areas (Figure 1), the quarries were sampled; a total of 186 cores from 36 sites was drilled. The clays are mostly blue-grey coloured; almost no digging was necessary to reach fresh material. Per site we collected 3 to 8 cylindrical standard cores using an electric drill.

In the laboratory, routine methods were applied in order to obtain characteristic magnetization directions for the clays. The Natural Remanent Magnetizations (NRM) were measured with a 2G-cryogenic magnetometer. The NRM of one sample per core was analyzed by means of progressive stepwise thermal demagnetization using small temperature increments: 40°C steps between 100 and 300°C and 10-30°C steps further up to 450-480°C. Individual NRM-measurements with intensities below 100 μA/m were performed twice to test the consistency of the measurement and (eventually) to increase precision by averaging the two independent measurements. During thermal treatment the low-field susceptibility (χ) of the samples was measured to detect changes in the magnetic mineralogy. At temperatures above 450°C the viscous behaviour relative to the remanence becomes too large to measure a meaningful direction.

Demagnetization diagrams were used for component

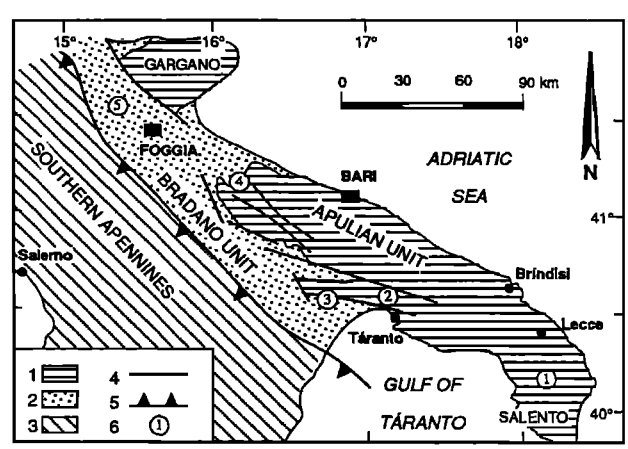


Figure 1. Sampled areas and major tectonic units around Apulia. 1=Apulia-Gargano foreland, 2=Plio-Pleistocene deposits, 3=Apenninic belt, 4=faults, 5=Apenninic front, 6=sampled locality (numbers: (1)=Cutrofiano, (2)=Monte-Mésola, (3)=Ginosa, (4)=Canosa di Púglia, (5)=Lucera).

analysis of each sample. For the low intensity sites, stacked demagnetization diagrams [Scheepers and Zijderveld, 1992] were used. The NRM-components were determined by calculating best fitting lines through data-points belonging to the temperature intervals, where minima in α_{95} of the stacked sum vector and/or the stacked difference vector exist. These magnetization vectors were combined using Fisher [1953] statistics to calculate mean directions per site and afterwards per quarry and per locality.

Results

The paleomagnetic results from Apulia will be presented going from the most southern part, the Salento peninsula, to the most northern part, the Gargano promontory.

Cutrofiano: On the Salento peninsula (Figure 1), we sampled early Pleistocene clays [formazione di Gallipoli, foglio 214, Carta geologica d'Italia] from two quarries south of Cutrofiano (Lecce district); in both quarries, CUT(A) and CUT(B), we drilled 3 sites. Bedding is sub-horizontal.

The samples have maximum blocking temperatures between 450 and 480°C (Figure 2A). The linear decrease to the origin from 360°C onwards was used to calculate a high-temperature (HT) component (Figure 2A); it has a normal polarity for all sites. This HT-component has a very good clustering per site (table 1); stacking confirms that the magnetization vector removed in this temperature trajectory (360-450°C) is the most consistent one in direction. The mean HT-direction for both quarries, using the site-means (table 2), implies that these clays do not show any tectonic rotation. Mean inclination (59.1°) is near the inclination (57.5°) of the geocentric axial dipole (GAD) field for the present latitude of the locality. All samples with a major decrease in NRM between 250 and 300°C also have a decrease in χ in this same trajectory. The trajectory for which the HT-component is demagnetized shows an increase in χ. Still, it was possible to measure the remanences with high consistency; after stacking the data a minimum in α_{95} in this trajectory is observed and a linear decrease to the origin as well.

MonteMésola: North of Táranto (Figure 1), we sampled three quarries in marine clays of late Pliocene - early Pleistocene age [Argilla del Bradano, foglio 202, Carta geologica d'Italia] east of the village of MonteMésola. In the largest and active quarry MM(A), we sampled 7 sites over a stratigraphic interval of 25 m. Only the upper two sites (MM 6-7)

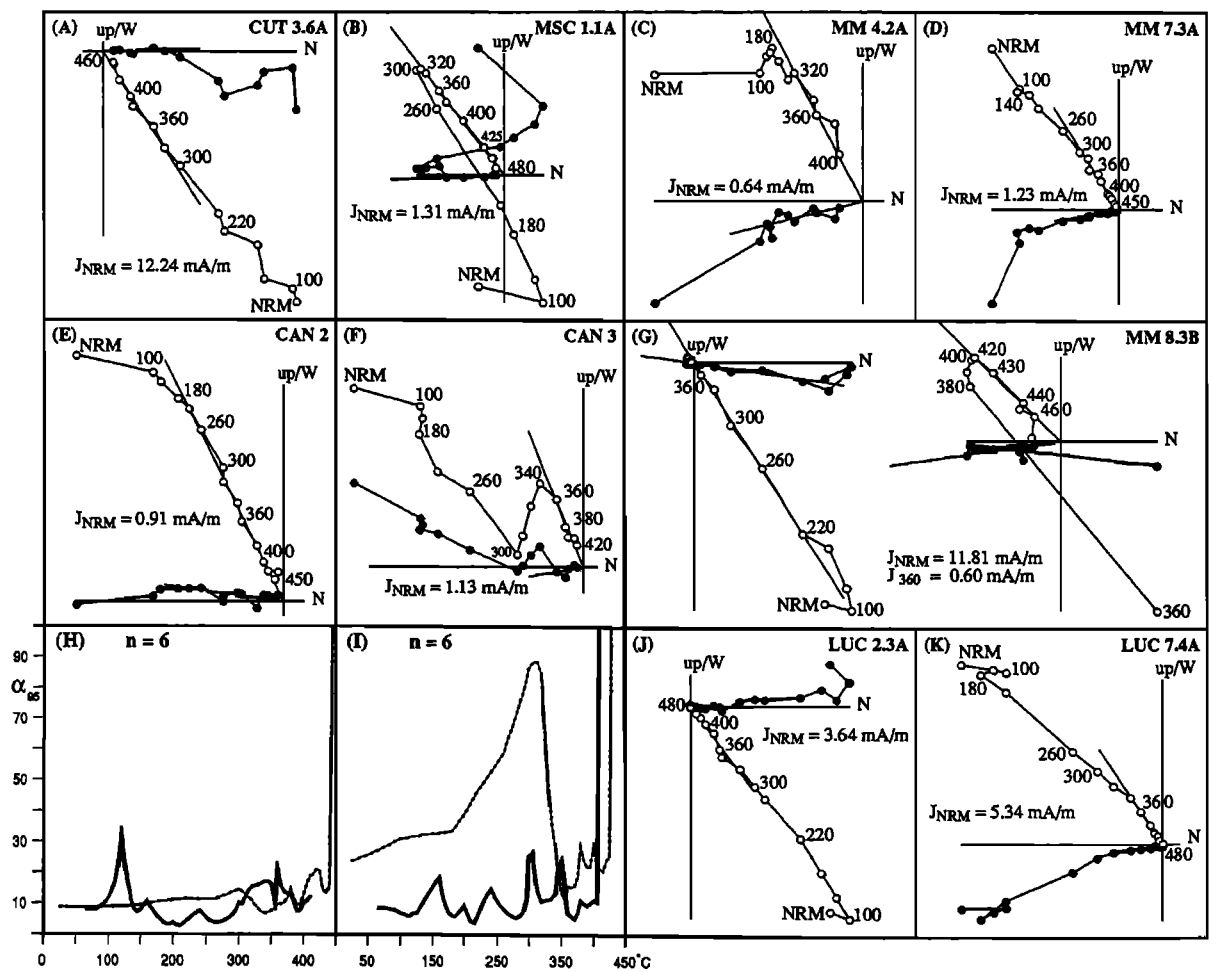


Figure 2. Stepwise thermal demagnetization diagrams of selected samples from Apulian clays. Numbers denote demagnetization steps in °C. Diagrams (E-F) represent stacked diagrams, diagrams (H-I) give variation of α_{95} for stacked sum vectors (dashed line) and stacked difference vectors (solid line) as a function of demagnetization temperature. The difference vectors are determined for 40°C temperature intervals; 'n' gives number of samples used for stacking.

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Table 1. Mean HT-directions for the sites from this study. N=number of samples, D(tc)=declination, I(tc)=inclination, k=precision parameter, α95=semi-angle of cone of confidence on the 95% level, mad=maximum angular deviation (stacked sites), Rs=length of resultant vector.

N	D(tc)	I (tc)	k	α95	pol	locality
Mean directions quarries:						
3 sites	2.6	59.1	1748	2.9	N	quarry CUT(A)
3 sites	-2.2	59.1	4383	1.9	N	quarry CUT(B)
7 sites	-3.9	53.5	67	7.4	R	quarry MM(A)
3 sites	-5.1	49.3	106	12.0	R	quarry MM(B)
3 sites	-10.6	55.2	75	14.3	R	quarry MM(C)
3 sites	2.8	57.1	264	7.6	R	quarry MSC(A)
4 sites	2.7	65.4	228	6.1	R	quarry CAN(A)
2 sites	1.6	52.6	Rs	1.937	R	quarry CAN(B)
3 sites	-0.1	56.6	279	7.4	N	quarry LUC(A)
4 sites	0.8	55.5	252	5.8	R	quarry LUC(B)
5 samples	1.9	48.8	88	8.2	R	quarry LUC(C)
Mean directions areas:						
2 quarries	0.2	59.1	Rs	2.000	N	Cutrofiano (LE)
3 quarries	-6.4	52.7	478	5.6	R	MonteMésola (TA)
1 quarry	2.8	57.1	Rs	1.000	R	Ginosa (MT)
2 quarries	2.0	59.0	Rs	1.988	R	Canosa (BA)
3 quarries	0.9	53.6	361	6.5	R-N	Lucera (FG)
Overall mean directions:						
11 quarries	-1.0	55.7	240	3.0	R-N	Apulian Unit
5 areas	-0.2	56.3	491	3.5	R-N	Apulian Unit
Geomagnetic observations:						
3 quarries	0.1	58.3	1800	2.9	N	polarity
8 quarries	-1.4	54.8	188	4.1	R	polarity

are in blue coloured clays; the other sites (MM 1-5) are in more green coloured clays. Bedding tilt is very slight (strike/dip: 204/02). The second quarry MM(B), situated east of the former and inactive, is stratigraphically equivalent of the upper part of quarry MM(A); we drilled 3 sites distributed over 5 m in blue coloured clays. Bedding tilt is slightly to the north (270/05). The third quarry MM(C) is active and to the north of the two others; we drilled 3 sites distributed over 5 m in blue coloured clays; bedding is sub-horizontal.

For the lowest 3 sites (MM 1-3) from quarry MM(A), initial NRM-intensities are only between 40 and 150 μA/m; the other sites have NRM-intensities above 250 μA/m, resulting in a well-defined demagnetization behaviour. In spite of these differences in NRM-intensity, the lowest 6 sites from quarry MM(A) give uniform final HT-directions (Figure 2C, D); the differences in NRM-intensity explain the differences of α_{95} per site. Site MM 7 has the most clear results (Figure 2D); initial intensity is above 1000 μA/m. The maximum blocking temperature is 450°C; the HT-component has a reversed polarity and a small CCW deviation from north-south (NS). The weaker sites of the lower part of the section show the same characteristics (Figure 2C); the deviation of declination is in the same order of magnitude; blocking temperature is less clear because above 380°C, the contribution of viscous magnetizations and scatter relative to the signal of interest becomes too large. The trajectory where the α_{95} of the vector sum for a site has its minimum value, was used to calculate the HT-direction.

All other sites of the MonteMésola area have a very dominating high-intensity (12-60 mA/m) normal polarity magnetization removed between 100 and 400°C (Figure 2G). This LT-component has no deviation in declination from NS

Table 2. Summary of main results from the paleomagnetic study in the clays overlying the Apulian Unit. pol=polarity (N=normal, R=reversed). Other symbols see table 1.

site	N	D(tc)	I(tc)	k	α95	site	N	D(tc)	I(tc)	k	α95
CUT 1	5	6.0	58.0	70	9.2	MM 10	3	184.3	-50.7	10	41.0
CUT 2	4	2.0	59.4	790	3.3	MM 11	4	176.5	-56.2	27	18.1
CUT 3	6	359.5	59.8	397	3.4	MM 12	4	156.8	-47.9	47	13.6
CUT 4	6	357.8	58.1	232	4.4	MM 13	4	178.3	-60.2	45	13.8
CUT 5	6	358.7	60.3	1496	1.7	CAN 1	7	177.5	-68.2	mad	5.2
CUT 6	6	356.8	58.8	841	2.3	CAN 2	6	188.0	-63.8	mad	4.3
MSC 1	6	178.0	-55.7	451	3.2	CAN 3	6	170.3	-67.8	mad	5.7
MSC 2	6	177.9	-56.5	75	7.8	CAN 4	5	191.5	-60.8	mad	6.5
MSC 3	6	193.3	-58.5	72	7.9	CAN 5	4	175.8	-38.8	mad	12.9
MM 1	3	168.9	-45.8	70	14.8	CAN 6	6	192.8	-65.9	mad	3.6
MM 2	4	201.9	-56.8	40	14.7	LUC 1	6	357.6	56.0	742	2.5
MM 3	4	181.4	-55.3	121	8.4	LUC 2	6	358.1	52.5	555	2.8
MM 4	4	185.7	-58.4	26	18.3	LUC 3	6	5.0	61.1	576	2.8
MM 5	4	177.8	-52.6	358	4.9	LUC 4	6	184.2	-62.4	96	6.9
MM 6	2	154.7	-49.1	Rs	1.969	LUC 5	6	179.6	-54.5	150	5.5
MM 7	3	169.9	-50.7	156	9.9	LUC 6	6	176.4	-55.8	85	7.3
MM 8	4	178.9	-49.9	103	9.1	LUC 7	5	182.2	-55.9	206	5.4
MM 9	4	162.5	-46.2	13	26.3	LUC 8	5	181.9	-48.8	88	8.2

(D_m=1.6°). For the sites showing this dominating LT-component, the final HT-component removed above 400°C and up to 460-470°C (Figure 2G), has a reversed polarity magnetization just like the other sites in this area. Above 450°C, the demagnetization is disturbed because the contribution of viscous magnetizations and because scatter relative to the signal of interest becomes too large. The intensity of the HT-component is between 70 and 920 μA/m; this is only in the order of one percent of the total NRM. The direction of the HT-component for these sites is the same as for the sites with no dominating normal polarity LT-component. The mean direction for the HT-component of the three quarries shows a small CCW-deviation from NS (D_m=173.6°). Mean inclination (52.7°) is slightly below the inclination (57.5°) of the GAD field for the present latitude of the locality.

The demagnetization behaviour of the sites in the MonteMésola area support the idea that the magnetization which is removed above 380-400°C is the primary magnetization for these clays. The direction of the HT-magnetization is not affected by (younger) LT-components. Again, there is the observation that for all sites χ increases during removal of the HT-magnetization without disturbing the NRM.

Ginosa: In the southeast of the Bradano Unit at its most external part (Figure 1), we sampled 3 sites west of the village Ginosa in an inactive quarry in blue coloured horizontally bedded clays [Argille subappennine, foglio 201, Carta geologica d'Italia]. The quarry is along the road to Montescagliose just east of the carbonate rocks of the Apulian Unit.

For the Ginosa-sites there is a distinction in two NRM-components (see Figure 2B). A normal polarity LT-component is removed for all 3 sites; only the temperature trajectory is different: 100-320° for site MSC 1 and 180-220° for sites MSC 2-3. The final decay is the same for all sites; up to 450-480°C a reversed polarity HT-component is removed. The mean HT-direction of the 3 sites does not show a significant deviation from NS (D_m=182.8°). Mean inclination (57.1°) is nearly the same as the inclination of the GAD field (57.5°) for the present latitude of the locality.

Canosa di Púglia: Near Canosa di Púglia (Bari district), we sampled two quarries in late Pliocene to early Pleistocene clays [Argille subappennine, foglio 176, Carta geologica d'Italia]. One quarry CAN(A) is just north of the SS.93 road;

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we drilled 4 sites over an interval of 15 m. The other quarry CAN(B) is just south of the SS.93 road; here we sampled 2 sites; the stratigraphic separation of the sites is only 2½ m.

For quarry CAN(A) thermal demagnetization gives irregular behaviour per core; therefore stacking of the signal [Scheepers and Zijderveld, 1992] was performed per site (Figure 2E, F). The lower 2 sites (CAN 3-4) show a clear minimum in the vector sum around 350°C (Figure 2I); therefore the HT-magnetization was calculated from this temperature onwards (up to 400°C) including the origin. For the upper 2 sites (CAN 1-2), the same temperature-trajectory was used to calculate the HT-magnetization (Figure 2E). The mean direction of the 4 sites is: D_m=182.7°and I_m= -65.4°. There is nearly no change in χ up to 380°C; from this temperature onwards χ starts to increase.

The results from quarry CAN(B) are also diffuse. After stacking the vector sum shows a minimum at 350°C for site CAN 5 and at 300°C for site CAN 6; therefore the calculation of the HT-component is from these temperatures onwards using the origin. The mean direction of the two sites is in the same order as for quarry CAN(A) (table 2). The sediments near Canosa di Púglia do not reflect any tectonic rotations either; for the 6 sites D_m=182.2° and mean inclination (61.2°) is somewhat steeper than the inclination (57.5°) of the GAD field for the present latitude of the locality.

Lucera: The most northerly area of this study is near Lucera (Foggia district) south-west of the Gargano peninsula (Figure 1) in the most external part of the Bradano Unit. Distributed over 3 (active) quarries we drilled 8 sites in very fresh rather silty clays [foglio 163, Carta geologica d'Italia]. Two of the quarries LUC(A) and LUC(B) are 10 km south of Lucera. Quarry LUC(A) is along the road SS.160, where we took sites LUC 4-7 from a stratigraphical interval of 7 m (strike/dip: 343/05). The other quarry LUC(B) to the south of Lucera, where we took sites LUC 1-3, was to the east of the first one; we sampled only 6 m (010/05). Site LUC 8 was in a quarry LUC(C) just north of Lucera (334/08).

Normal and reversed polarities are present in the Lucera area. Here as well the samples have lost their magnetization between 450 and 480°C (Figure 2J, K). The sites in quarries LUC(A) and LUC(C) have a reversed polarity HT-magnetization; the sites in quarry LUC(B) have a normal polarity HT-magnetization. The mean HT-direction for the three quarries is: D_m= 0.9°and I_m= 53.6° (table 2).

Averaging the 5 sampled areas results in a mean direction for the young clays covering the Apulia-Gargano foreland, implying no rotation since the Pleistocene: D_m=359.8° and I_m= 56.3° (table 2). Averaging the 11 quarries gives a similar result: D_m=359.0°and I_m= 55.7°. The mean direction for the 3 quarries with a normal polarity HT-component is: D_m= 0.1° and I_m= 58.3° (k=1800); for the 8 quarries with a reversed polarity: D_m=178.6° and I_m= -54.8° (k=188). The angle between the two directions is 176.4°. The reversal test ('Different Kappa' test) is positive, classification 'A' [McFadden and McElhinny, 1990].

Conclusions

The HT-component of the remanence carried by these marine marly clays reflects the earth's magnetic field. The HT-component is present in all sites, even in (1) sites with weak NRM, in (2) sites with dominating LT-components and in (3) sites with irregular behaviour per core. Because of the very low tilt we don't have the opportunity to perform fold-tests for the HT-component. Stacking shows that per site the component removed between 360 and 480°C, is the most consistent one. Both normal and reversed polarities are present for the HT-component; the positive reversal test supports the stability of the paleomagnetic HT-directions. Since the sites have a late Pliocene to early Pleistocene age, the reversed polarity HT-components must reflect the Matuyama Chron; the normal polarity HT-component in the two quarries near Cutrofiano and quarry LUC(A) near Lucera probably represents the Jaramillo (or Olduvai) subchron. Further there is no significant deviation from the inclination of the geocentric axial dipole field for the present latitude of the locality (56.3 versus 57.5°). The conclusion is that there is no rotation for the Apulia-Gargano foreland since the Pleistocene. This implies that the rotation which is present in the Cretaceous limestones from the Gargano peninsula, has occurred earlier in time than the Pleistocene rotation of the southern Apennines itself.

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